PROJECT TITLE

For

Sensor Lab mini Project

Third Year (Semester-VI) Bachelors in Information Technology, Autonomous Program

By

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Certificate

This is to certify that have completed the project report on the topic satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Dr. Monali Chaudhari during the year 2023-2024 as prescribed by An Autonomous Institute Affiliated to University of Mumbai

Supervisor/ Examiner

Abstract

This project involves designing and implementing a Contactless Automatic Water Dispenser using an Arduino Uno, Ultrasonic Sensor, Servo Motor, and basic electronic components. The dispenser operates autonomously by detecting the presence of a user's hand and activating the servo motor to release water without physical contact. This is achieved through the integration of distance sensing, motor control, and simple automation logic programmed using Arduino. The primary objective is to demonstrate the practical application of sensor integration, embedded systems, and touchless technology. This project serves as a fundamental example of how microcontrollers can be used to automate hygienic systems, with potential applications in public sanitation, smart homes, and healthcare facilities.

Chapter 1: Introduction

A Contactless Automatic Water Dispenser is a hygienic solution designed to dispense water without physical contact by using an Arduino Uno, Ultrasonic Sensor, and Servo Motor. The Arduino Uno acts as the central controller, processing distance data from the ultrasonic sensor, which emits sound waves and measures their reflection to detect the presence of a user's hand. The sensor continuously monitors the area in front of the dispenser, and when an object is detected within a specific range, the Arduino activates the servo motor to open the water valve and allow water to flow. After a short delay, the servo returns to its original position, stopping the water flow. This automatic process ensures a touch-free experience, reducing the risk of contamination and promoting hygiene. The system is simple, reliable, and cost-effective, making it ideal for deployment in public places, homes, schools, and healthcare facilities where touchless operation is essential.

Chapter 2: Literature Review

Survey

Various contactless dispensing and automation systems utilize sensors such as Ultrasonic, Infrared (IR), and Capacitive sensors. Among these, Ultrasonic Sensors are preferred due to their reliability, affordability, and ease of integration with microcontrollers like Arduino. These systems are commonly implemented in public hygiene applications to reduce physical contact and prevent the spread of infections.

Research Gap

Most contactless dispensing systems available in the market are either costly, complex, or not easily accessible for educational or small-scale implementations. This project aims to develop a simple, low-cost, and efficient automatic water dispenser using basic components, making it suitable for students, hobbyists, and community installations.

Related Work

1. "Smart Water Dispenser Using Ultrasonic Sensor and Arduino" by R. Mehta and K. Sharma(2020): This paper presents the development of a contactless water dispenser using ultrasonic sensors and Arduino, emphasizing the importance of hygienic water access and non-contact operation in public settings.
2. "Automatic Touchless Water Dispenser System" by M. Verma and N. Rao (2019): The study outlines a system based on ultrasonic sensing and servo motor control to provide hands-free water dispensing. The paper highlights improvements in hygiene and user convenience through simple sensor-based automation.

Chapter 3: Design Procedure

Problem Definition

The objective of this project is to design and develop a low-cost, efficient Contactless Automatic Water Dispenser using an Arduino Uno and commonly available components such as an Ultrasonic Sensor, Servo Motor, and basic circuitry. This project aims to demonstrate fundamental automation principles such as proximity sensing, motor control, and hands-free operation using simple electronic components and Arduino programming.

3.2 Steps Involved

1. **Component Selection and Procurement**
   - Selecting appropriate components required for building the contactless automatic water dispenser, including:
     - Arduino Uno (Microcontroller for processing sensor data and controlling motors)
     - Ultrasonic Sensor (HC-SR04) (For detecting obstacles by measuring the distance from objects using sound waves)
     - Servo Motor (To rotate the ultrasonic sensor and enable wider scanning angles)
     - Breadboard and Jumper Wires (For assembling and testing the circuit without soldering)
     - Power Supply (Batteries or External Source) (To power the entire system)
   - Procuring these components from reliable sources to ensure compatibility and proper functioning.

2. **Circuit Assembly and Wiring**
   - Assembling the dispenser by setting up all components on a breadboard for easy prototyping.
   - Connecting the Ultrasonic Sensor to the Arduino’s digital pins for sending and receiving ultrasonic signals.
   - Connecting the Servo Motor to one of the Arduino’s PWM pins to control angular movement.
   - Ensuring proper grounding and avoiding loose connections to maintain system stability.

3. **Arduino Programming**

    - Writing a program using the Arduino IDE to control the entire system.
    - Programming the Arduino to:
        - Continuously monitor the distance measured by the ultrasonic sensor.
        - Trigger the servo motor to rotate and open the valve when a hand is detected within a set range.
        - Introduce a delay for water flow and then return the servo to its original position to stop dispensing.
        - Ensure responsive and accurate activation for a smooth, contactless user experience.

4. **Testing and Calibration**

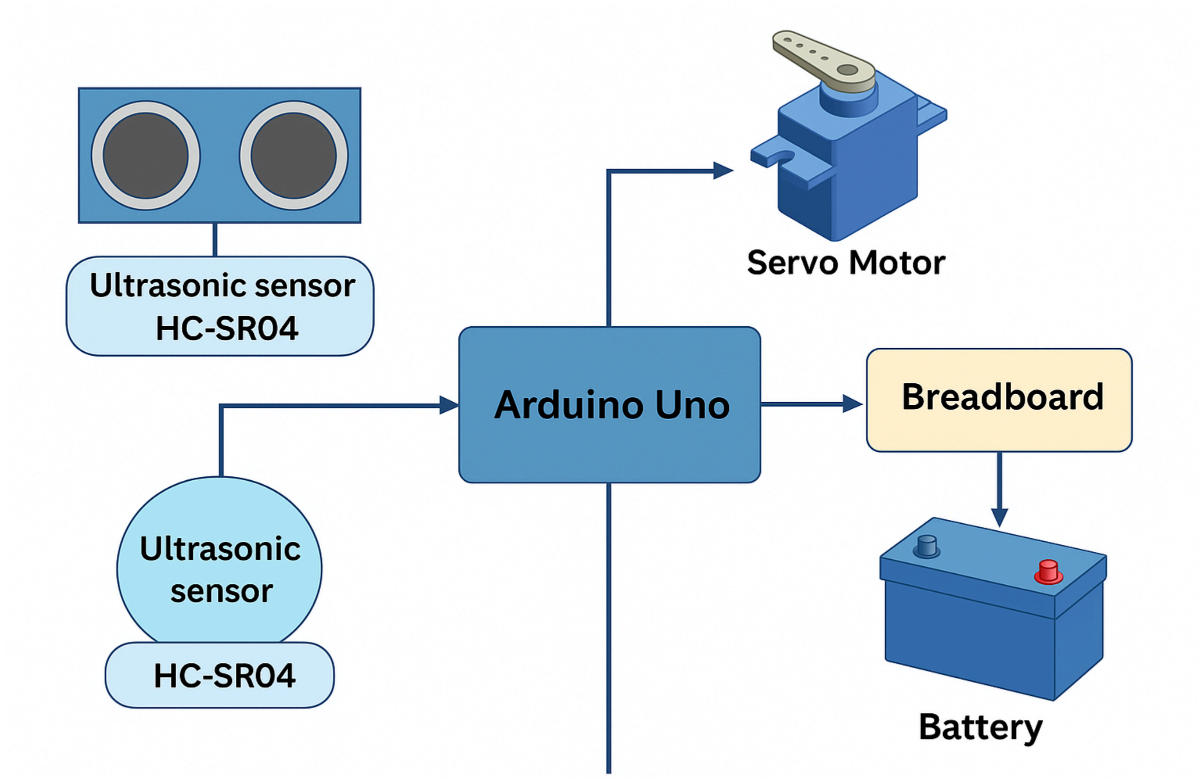
    - Testing the system's response to hand movement and ensuring accurate water dispensing.
    - Calibrating the Ultrasonic Sensor by adjusting distance threshold values in the code for optimal hand detection.
    - Testing the Servo Motor positions to ensure proper valve operation.
    - Fine-tuning delay timings and distance sensitivity to reduce false triggers and enhance usability.

5. **Final Assembly and Demonstration**

    - Securing all components properly on a stable platform after successful testing.
    - Demonstrating the contactless water dispensing by placing a hand within range and observing automatic water flow.

- Evaluating system responsiveness, reliability, and effectiveness in a real-world hygiene application.

3.3 Block Diagram of Proposed Project



3.4 Component Description

Hardware Components

1. **Arduino Uno -** Controls the entire system and processes sensor data.
2. **Ultrasonic Sensor (HC-SR04)** - Measures distance to obstacles.
3. **Servo Motor** - Rotates ultrasonic sensor for directional scanning.
4. **Connecting Wires** - Used for establishing connections.

Specifications

- **Arduino Uno:** 5V, 14 Digital I/O Pins, 6 Analog Inputs, Clock Speed: 16 MHz.
- **HC-SR04:** Operating Voltage: 5V, Range: 2cm - 400cm, Accuracy: ±3mm.
- **Servo Motor:** Operating Voltage: 4.8V-6V, Angle Range: 0°-180°.

3.5 Working of Proposed Project

- The Ultrasonic Sensor continuously emits sound waves and measures the time taken for the echo to return.
- The Arduino Uno calculates the distance between the sensor and a nearby object (such as a user's hand) based on the speed of sound.
- If an object is detected within a predefined threshold distance, the Arduino triggers the Servo Motor.
- The Servo Motor rotates to open the water valve, allowing water to flow.
- After a short delay, the Servo Motor returns to its original position to stop the water flow.

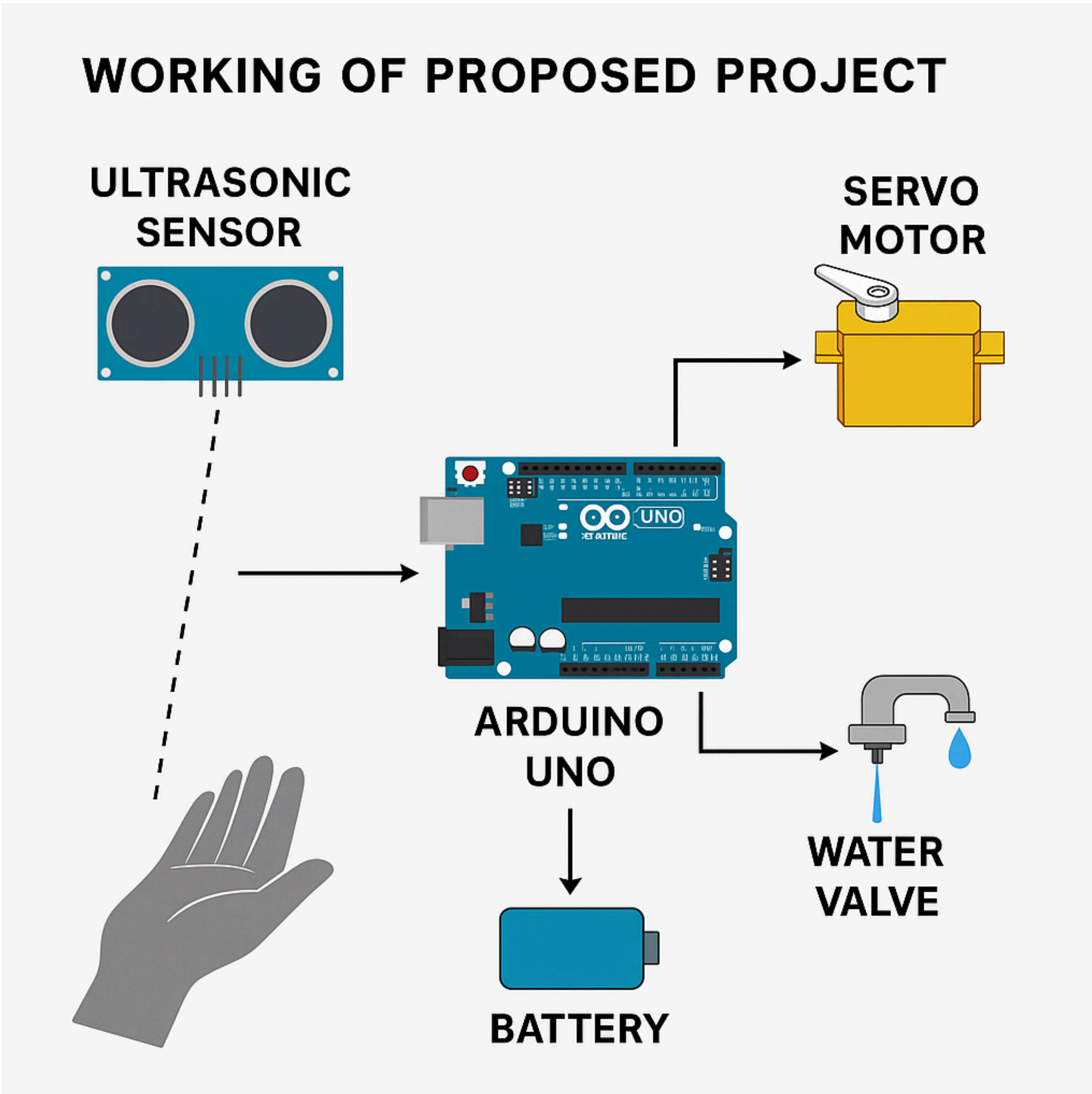
Chapter 4: Implementation

Hardware Implementation

- Connections between Arduino, Ultrasonic Sensor and Servo Motor.
- Circuit diagram and wiring details.

Software Implementation

Flowchart :



- Arduino programming using **C/C++ (Arduino IDE)**.

Code

```
#include <Servo.h>
#define trigPin 10
#define echoPin 11
#define servoPin 9

Servo waterServo;
void setup() {
  Serial.begin(9600);

waterServo.attach(ser
voPin);
  pinMode(trigPin,
OUTPUT);
  pinMode(echoPin,
INPUT);
  waterServo.write(0);
}

void loop() {
  float distance =
measureDistance();
  if (distance > 0 &&
distance < 10) {

waterServo.write(90);
        delay(1000);

waterServo.write(0);
        delay(1000);
  }
}
```

Chapter 5: Results and Discussion

Simulation Output

The simulation of the contactless automatic water dispenser was successfully carried out using the Arduino IDE. The ultrasonic sensor accurately measured the distance of approaching objects, and the servo motor responded effectively by rotating to simulate water dispensing. The programmed logic reliably triggered the servo when a hand was detected within the set threshold distance, ensuring proper automation in a simulated environment.

Actual Output

In real-world testing, the physical setup of the water dispenser performed as expected. The ultrasonic sensor consistently detected hand proximity, and the servo motor operated smoothly to open and close the valve for water flow. The system provided a completely touch-free user experience, validating its effectiveness for hygiene-focused applications.

Discussion

The project successfully demonstrated efficient proximity detection and hands-free water dispensing. While the system met its core objectives, possible improvements include:

- Integrating a timer or flow sensor to optimize water usage.
- Enhancing the range detection logic to reduce false triggering in crowded environments.

Overall, the contactless water dispenser functioned reliably, showing promise for use in homes, offices, and public facilities where hygiene and convenience are priorities.

References

1. **Arduino Documentation:** https://www.arduino.cc/reference/en/
2. **HC-SR04 Datasheet:** https://www.electroschematics.com/hc-sr04-datasheet/
3. **Servo Motor Datasheet**: https://www.towerpro.com.tw/product/sg90-9g-micro-servo/